

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

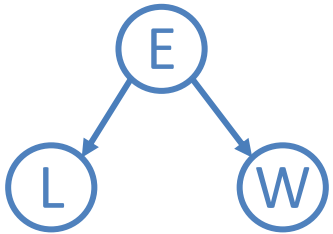
- Classwork 3
  - Available on Brightspace/GitHub Classroom
  - Due **Friday 9/27 @5 pm**
- Homework 1
  - Due Friday 10/4 @5 pm

This Lecture

- Bayesian Network
  - Inference Goals
  - d-Separation
  - Exact Inference

Last Lecture

- Markovian Factorization
  - Goal?
    - Reconstruct JPT only using the CPTs
  - How?
    - $P(V_1, V_2, \dots) = \prod_{V_i \in \boldsymbol{V}} P(V_i | parents(V_i))$
  - What is are the CPTs?
  - What is the JPT?



Probabilistic Inference with BN

- Input
  - **Query**: a query over network variables Q
  - **Evidence**: any observed evidence e
- Output
  - **Conditional distribution:** $P(Q|e)$
- Notation: $BN.ask(Q, e) = P(Q|e)$

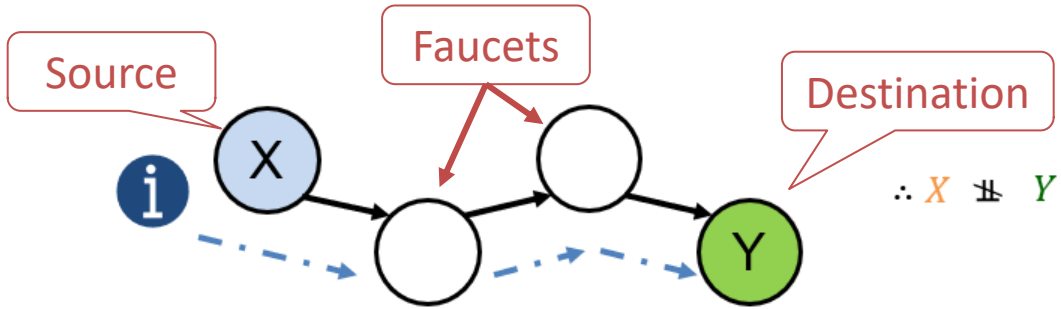
Inference Goal

Probabilistic Inference with BN

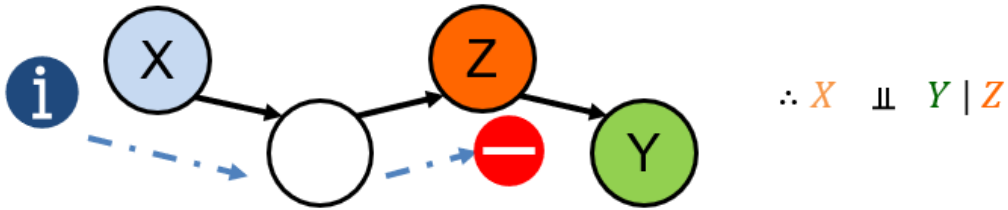
- Goal
  - A general inference procedure to compute P(Q|e) given a BN. What are available in a BN?
    - Structure (i.e., DAG)
    - Semantics (i.e., CPTs)
  - Determine the **independence** relationships (⫫) between variables from the structure of BN

Information Flow

- On / Open: **Dependent**



- Off / Closed/Blocked: **Independent**

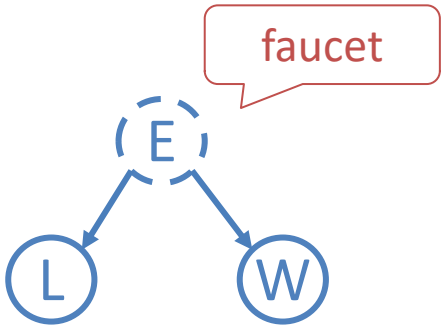


d-Separation

- **Directional-separation** (d-separation) determine a Bayesian Network's **independence and conditional independence** claims directly from the network's **structure** (i.e., directionality of edges in DAG).
- How?
  - Path Analysis
  - $dSep(X, Z, Y) \rightarrow T/F \Leftrightarrow X \perp\!\!\!\perp Y|Z$

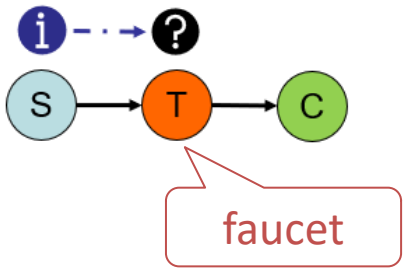
Rule 1: Forks $X \leftarrow Z \rightarrow Y$

- A **fork** $X \leftarrow Z \rightarrow Y$ is **blocked** along a path if Z is **given** (i.e., if a common cause is given, a fork is blocked). Otherwise, it is **open**.
- Example
  - When E is not given, L⫫W?
    - No, L⫫̸W (faucet is open)
  - When E is given, L⫫W|E?
    - Yes, L⫫W|E (faucet is closed)



Rule 2: Chains $X \rightarrow Z \rightarrow Y$ OR $X \leftarrow Z \leftarrow Y$

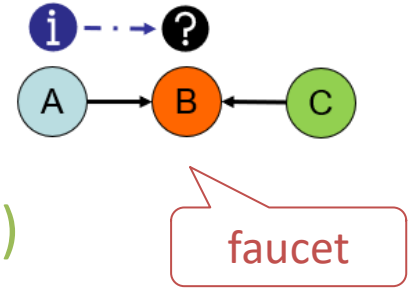
- A **chain** $X \rightarrow Z \rightarrow Y$ or $X \leftarrow Z \leftarrow Y$ is **blocked** along a path if Z is **given**.
- Example
  - When T is not given, S⫫C?
    - No, S⫫̸C (faucet is open)
  - When T is given, S⫫C | T?
    - Yes, S⫫C | T (faucet is closed)



S = a patient smokes
T = a patient has lung tar
C = a patient has lung cancer

Rule 3: Sinks $X \rightarrow Z \leftarrow Y$

- A **sink** $X \rightarrow Z \leftarrow Y$ is **blocked** along a path if **neither** Z **nor its descendants are given.**
- Example
  - When B is not given, A⫫C?
    - **Yes**, A⫫C (faucet is **closed**)
  - When B is given, A⫫C|B?
    - **No**, A⫫̸C|B (faucet is **open**)



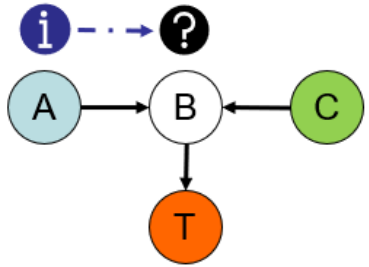
A = a fair coin flip

C = a fair coin flip

B = a bell that rings if both A and C are heads or both tails

Rule 3: Sinks $X \rightarrow Z \leftarrow Y$

- Example
  - Adding Descendants
  - When T is given, A⫫C | T?
    - **No**, A⫫̸C | B (faucet is **open**)



A = a fair coin flip

C = a fair coin flip

B = a bell that rings if both A and C are heads or both tails

T = Butlerbot 6000 brings us tea if the bell rings

d-Separation Algorithm

- Rules

Triplet Type	Format	Rule (Blocked if …)
Fork	$X \leftarrow Z \rightarrow Y$	Blocked if **Z is given**
Chain	$X \leftarrow Z \leftarrow Y$ $X \rightarrow Z \rightarrow Y$	Blocked if **Z is given**
Sink	$X \rightarrow Z \leftarrow Y$	Blocked if **neither Z nor its descendants are given**

d-Separation Algorithm

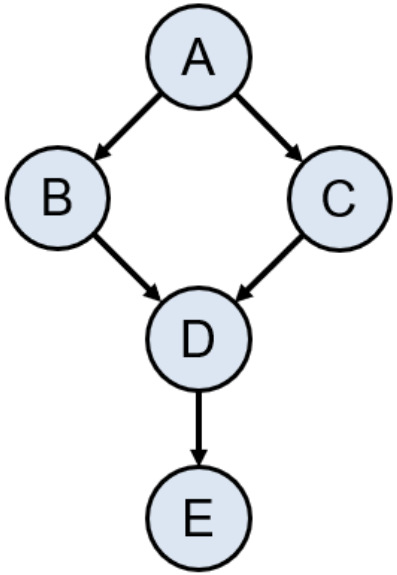
- By-Hand Version

```
# Returns true if X is independent of Y given Z, else false
def d_sep(X, Y, Z):
    for each undirected path p between X and Y:
        openPath = true
        for each triplet t along p:
            if t is blocked:
                openPath = false
                break
            if openPath:
                return false
    return true
```

Example

- A⫫D|B,C?
- A⫫D|B?
- C⫫B|A?
- C⫫B|A,E?

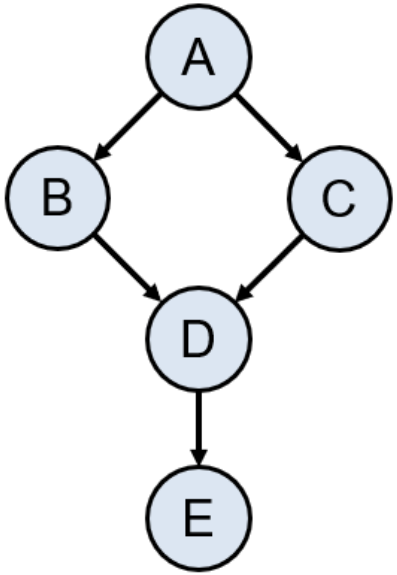
Triplet Type	Format	Rule (Blocked if …)
Fork	$X \leftarrow Z \rightarrow Y$	Blocked if **Z is given**
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Sink	$X \rightarrow Z \leftarrow Y$	Blocked if **neither Z nor its descendants are given**



Example

- A⫫D|B,C
  - $A \rightarrow B \rightarrow D$ Blocked Chain
  - $A \rightarrow C \rightarrow D$ Blocked Chain
- A⫫D|B
  - $A \rightarrow B \rightarrow D$ Blocked Chain
  - $A \rightarrow C \rightarrow D$ **Open** Chain

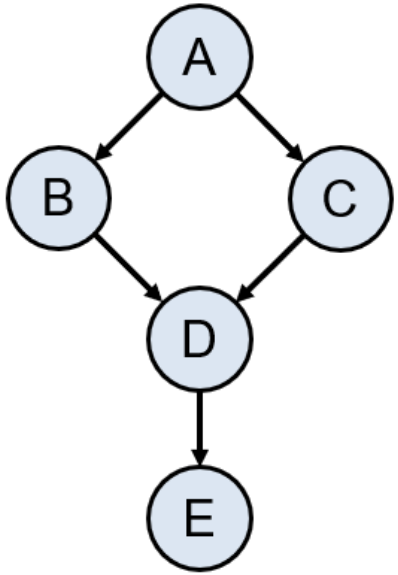
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Fork	$X \leftarrow Z \rightarrow Y$	Blocked if **Z is given**
Chain	$X \leftarrow Z \leftarrow Y$ $X \rightarrow Z \rightarrow Y$	Blocked if **Z is given**
Sink	$X \rightarrow Z \leftarrow Y$	Blocked if **neither Z nor its descendants are given**



Example

- C⫫B|A
  - $B \leftarrow A \rightarrow C$ Blocked Fork
- C⫫̸B|A,E
  - $B \leftarrow A \rightarrow C$ Blocked Fork
  - $B \rightarrow D \leftarrow C$ **Open** Sink
    - A descendent of D is given

Triplet Type	Format	Rule (Blocked if …)
Fork	$X \leftarrow Z \rightarrow Y$	Blocked if **Z is given**
Chain	$X \leftarrow Z \leftarrow Y$ $X \rightarrow Z \rightarrow Y$	Blocked if **Z is given**
Sink	$X \rightarrow Z \leftarrow Y$	Blocked if **neither Z nor its descendants are given**



Exact Inference in BN

- What is the goal of inference in BN?
  - $BN.ask(Q, e) = P(Q|e)$
- However, certain queries are easier to answer than others. Why?
  - Arbitrary queries of the format $P(Q\,|\,e)$ are not always immediately available from the BN.
- What are immediately available from the BN?
  - Structure (i.e., DAG) and Semantics (i.e., CPTs)

Query Difficulty

- O(1) Answerable Queries
  - Can be answered in $O(1)$ directly from one of the network's CPTs / parameters (without needing to perform any computation).
- Example: $X \rightarrow Z \rightarrow Y$
  - CPTs: P(X), P(Z|X), P(Y|Z)
  - Query: P(Z=1|X=0)
    - Answerable in O(1) by looking up in CPT on Z

What are the CPTs?

Query Difficulty

- O(|V|) Answerable Queries
  - **Linearly Answerable** O(|V|) Queries (for |V| variables) are those can be answered using ONLY Markovian Factorization.
  - A row in JPT that can be reconstructed from CPTs
- Example: $X \rightarrow Z \rightarrow Y$

What is the JPT given the CPTs ?

  - JPT: P(X, Z, Y) = P(X) P(Z|X) P(Y|Z)
  - Query: P(X=0, Z=1, Y=1)
    - = P(X=0) P(Z=1|X=0) P(Y=1|Z=1)

Query Difficulty

- O(|V|^2) Answerable Queries
  - **Quadratic Answerable** O(|V|^2) Queries are those for which application of d-separation, with cost O(|V|) , need be applied at most $|V|$ times to reduce a query to a O(1) format.
- Example: $X \leftarrow Z \rightarrow Y$
  - Query: P(X|Z,Y)
  - d-Sep: P(X|Z,Y) =P(X|Z) → O(1) query by looking up CPT on X

Example

- BN: $A \rightarrow B \rightarrow C$
- What is the difficulty of each query?
  - P(B|A) O(1)
  - P(C|A,B) $O(|V|^2)$
  - P(A|B) Cannot be obtained efficiently.
  - P(A,B,C) $O(|V|)$

Remaining of the lecture

HOMEWORK 1 / CLASSWORK 3